

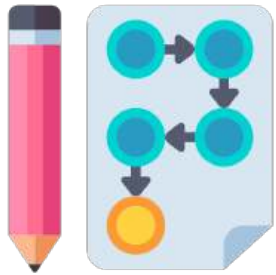
ISA CERTIFIED PERSONAL TRAINER COURSE

WEEK 5 – PROGRAM DESIGN 2 (CARDIORESPIRATORY TRAINING)

- CHAPTER 8
CARDIORESPIRATORY TRAINING: PHYSIOLOGY,
ASSESSMENTS AND PROGRAMMING
- CHAPTER 11
INTEGRATED EXERCISE PROGRAMMING:
FROM EVIDENCE TO PRACTICE



CONTENT COVERED

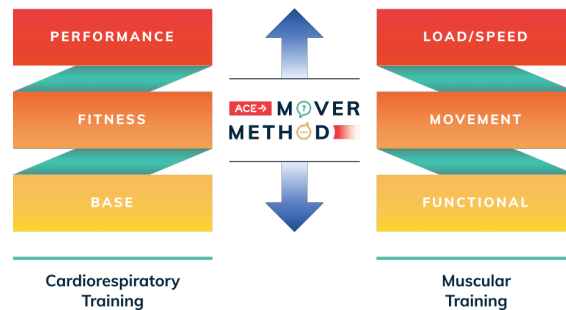


CT Program Design



Components of a CT Session

ACE→ Integrated Fitness Training® Model



ACE-IFT Model for CT



Evidence-Based Practice

CHAPTER 8
CARDIORESPIRATORY TRAINING:
PHYSIOLOGY, ASSESSMENTS AND
PROGRAMMING



CHAPTER OVERVIEW

1. What is cardiorespiratory/aerobic exercise? What is endurance?
2. Why cardio exercise?
3. What is the recommended training frequency and volume?
4. What are the steps of designing an CT program?
5. What are some typical modes/types of CT program?
6. What are the components of an CT session?
7. What are the methods of determining exercise zones?
8. What is the appropriate rate to increase intensity/volume?
9. How to design a CT program based on ACE-IFT model?

A close-up, low-angle shot of a person's legs and feet as they run on a green grassy field. The person is wearing blue and black patterned leggings and blue Nike sneakers with black swooshes and laces. The background is a blurred landscape with trees and a cloudy sky.

*WHAT IS
CARDIORESPIRATORY
TRAINING?*

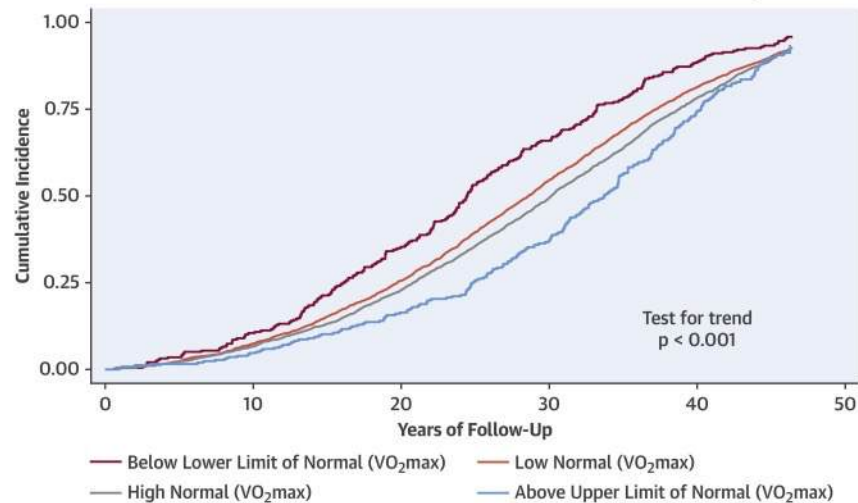


*WHY DO WE DO
CARDIO TRAINING?*

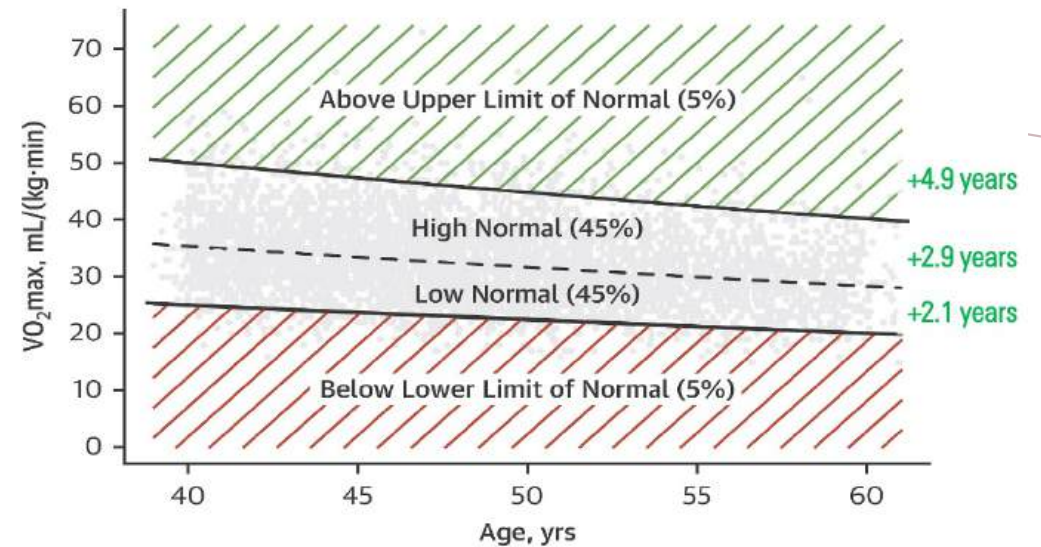
CENTRAL ILLUSTRATION: Physical Fitness and Longevity: All-cause Mortality

Midlife Cardiorespiratory Fitness Maximal Oxygen Consumption (VO_2max) & All-Cause Mortality

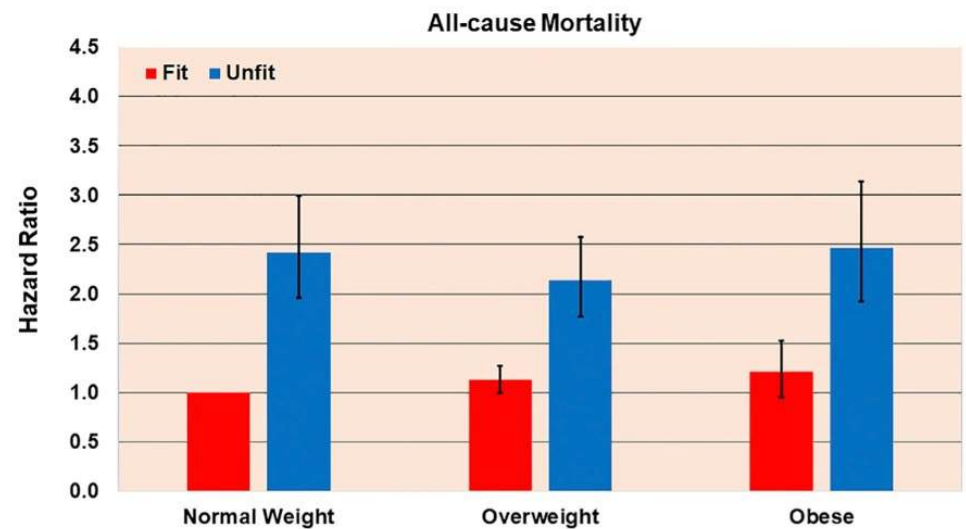
5,107 Men Free of Cardiovascular Disease at Inclusion Followed up to 46 Years



Clausen, J.S.R. et al. J Am Coll Cardiol. 2018;72(9):987-95.



Each unit increase in VO_2max ($1 \text{ mL/kg} \cdot \text{min}^{-1}$)
~ 45 days extension in life expectancy



CARDIORESPIRATORY TRAINING PROGRAM DESIGN

Learning Objectives:

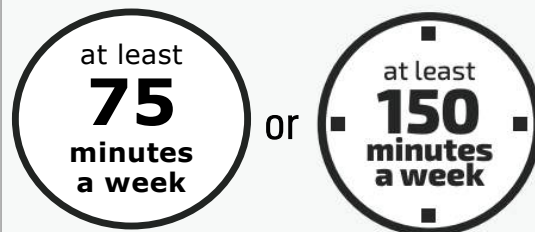
- Understand the general principles of physical training
- Identify key programming considerations for limiting sedentary time
- List the three phases of the ACE IFT Model for cardiorespiratory training
- Design personalized cardiorespiratory training programs for clients ranging from physically inactive individuals to endurance athletes utilizing assessment and programming tools from each phase of the ACE IFT Model

PHYSICAL ACTIVITY GUIDELINES

How much activity do I need?

Moderate-intensity aerobic activity

Anything that gets your heart beating faster counts.



vigorous

moderate



AND

Muscle-strengthening activity

Do activities that make your muscles work harder than usual.



Tight on time this week? **Start with just 5 minutes.** It all adds up!

Adapted from: <https://www.cdc.gov/physicalactivity/basics/adults/index.htm>

CT EVIDENCE-BASED RECOMMENDATIONS

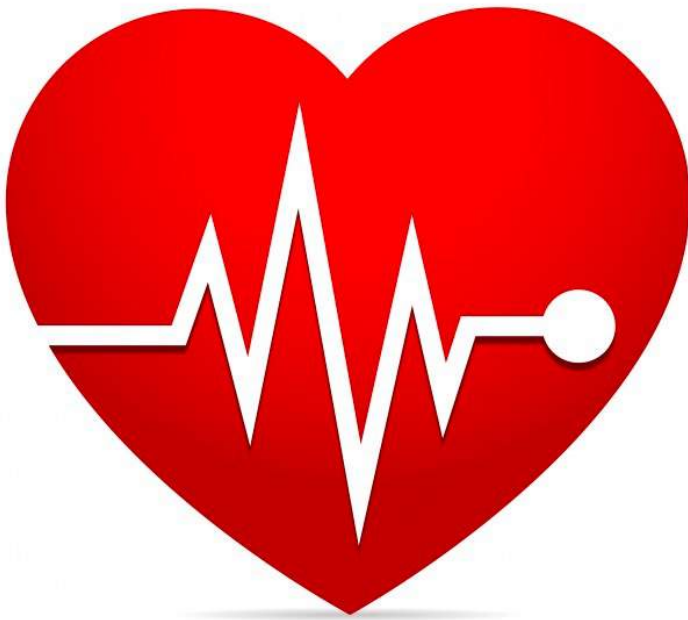
Frequency	$\geq 5\text{d/wk}$ (moderate), or $\geq 3\text{d/wk}$ (vigorous), or combo
Intensity	Moderate to vigorous for most adults. Light to moderate intensity for deconditioned individuals.
Time	30-60 min/d (moderate), or 20-60 min/d (vigorous), or combo <20 min/d can be beneficial for previously sedentary individuals
Type	Regular, purposeful, involving all major muscle groups and is continuous and rhythmic in nature
Volume	$\geq 500\text{-}1,000$ MET-min/wk. ≥ 150 min/wk (moderate) or $\geq 7,000$ steps/d. Smaller volume is still beneficial.
Pattern	Continuous or interval or cumulative of ≥ 10 min bouts. <10 min bouts may still benefit deconditioned individuals.
Progression	Start low and go slow. Increase the step count by $\geq 2,000/\text{d}$ until target volume reached.

INTENSITY MONITORING METHODS

- Heart rate [% maximal heart rate (MHR); % heart-rate reserve (HRR)]
- Ratings of perceived exertion (RPE)
- VO_2 or metabolic equivalents (METs)
- Caloric expenditure
- Talk test/first ventilatory threshold (VT1)
- Blood lactate and second ventilatory threshold (VT2)

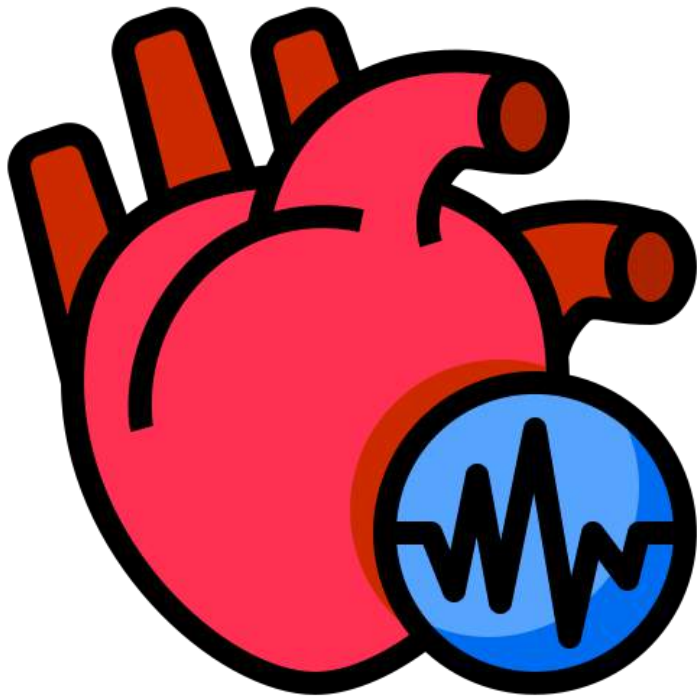


INTENSITY - HEART RATE



- % of MHR or HRR: most widely used for programming and monitoring intensity
- Variables impacting MHR:
 - Genetics
 - Exercise modality (e.g., MHR varies between running and cycling)
 - Medications
 - Body size
 - Altitude – lowers MHR reached due to a client's inability to train at higher intensities
 - Age – MHR varies significantly among people of the same age

AGE-PREDICTED MHR (APMHR)



- Tends to overestimate MHR in younger adults
- Underestimates MHR in older adults

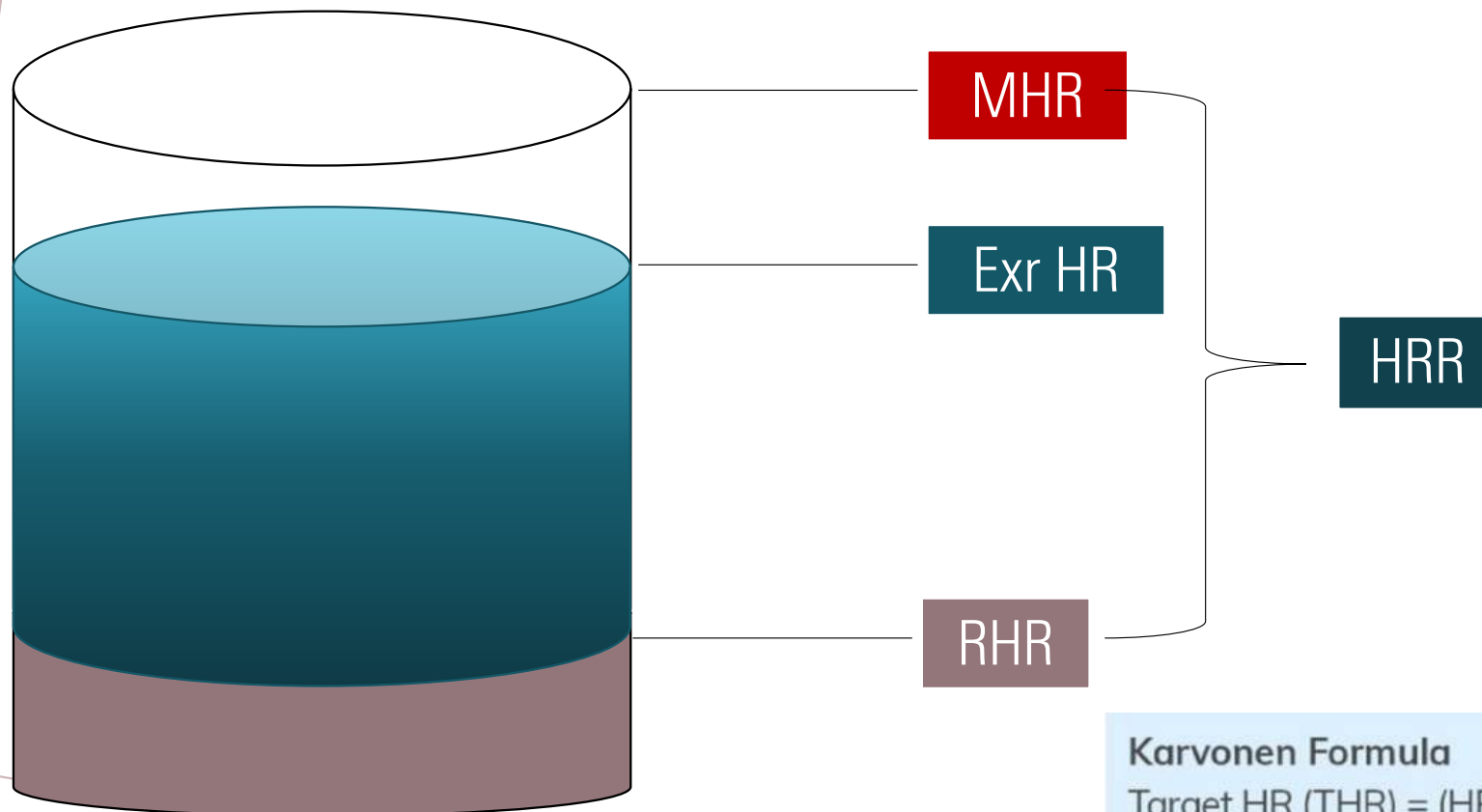
Formula:

$$\text{MHR} = 220 - \text{Age}$$

Formulas exist with standard deviations closer to 7 bpm:

$$\begin{aligned} &206.9 - (0.67 \times \text{age}) \\ &208 - (0.7 \times \text{age}) \end{aligned}$$

KARVONEN METHOD



Legend:

MHR → Maximal Heart Rate

Ex HR → Exercise Heart Rate

RHR → Resting Heart Rate

HRR → Heart Rate Reserve

Karvonen Formula

Target HR (THR) = (HRR x % Intensity) + RHR

Where: HRR = MHR – RHR

THREE-ZONE TRAINING MODEL



ZONE 1
40-60% HRR

ZONE 2
60-90% HRR

ZONE 3
90-100% HRR

Refer to Table 8-11: Three-zone Intensity Model Using Various Intensity Markers

INTENSITY - RPE

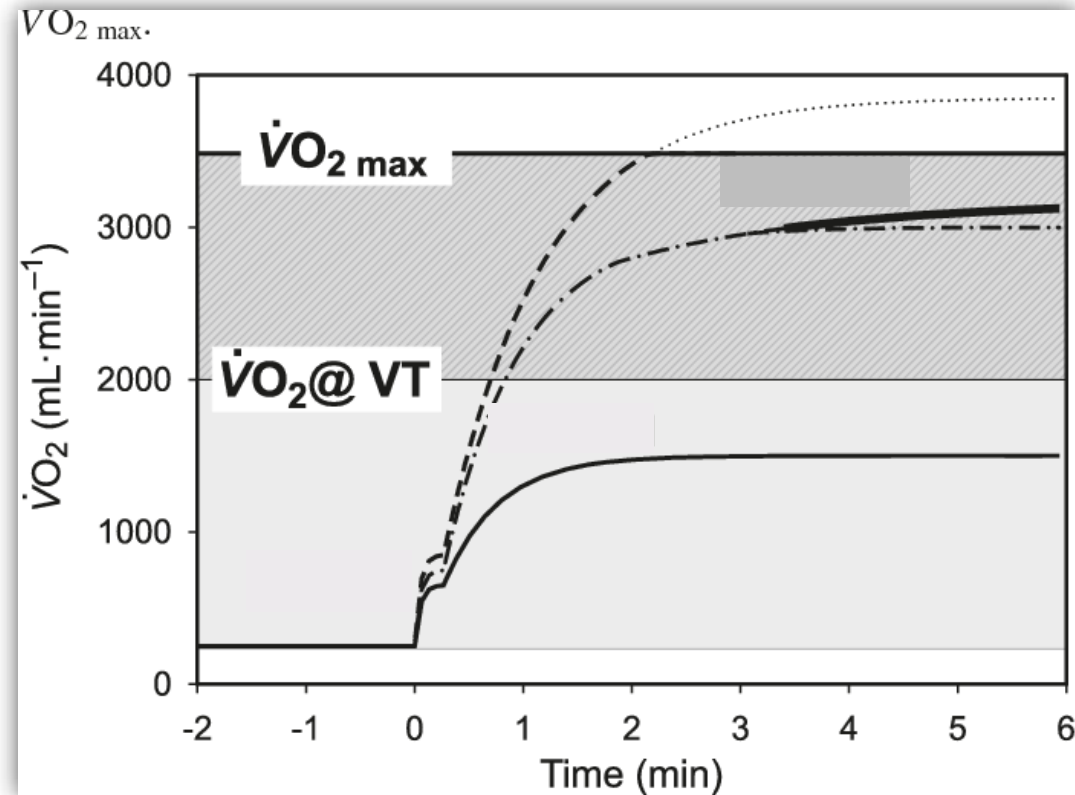
6	At rest	0	At rest
7	Very, very light	1	Very easy
8		2	Somewhat easy
9	Very light	3	Moderate
10		4	Somewhat hard
11	Fairly light	5	Hard
12		6	
13	Somewhat hard	7	Very hard
14		8	
15	Hard	9	Very, very hard
16		10	Maximum
17	Very hard		
18			
19	Very, very hard		
20	Maximum		



INTENSITY - $\dot{V}O_2$ AND METS

[PG 282-283]

- $\dot{V}O_2$: oxygen uptake
- $\dot{V}O_2$ (at rest) $\approx$ 3.5 mL/kg/min
Intensity: % $\dot{V}O_{2\max}$ or % $\dot{V}O_{2R}$
- 1 MET = 3.5 mL/kg/min
Intensity: xMETS



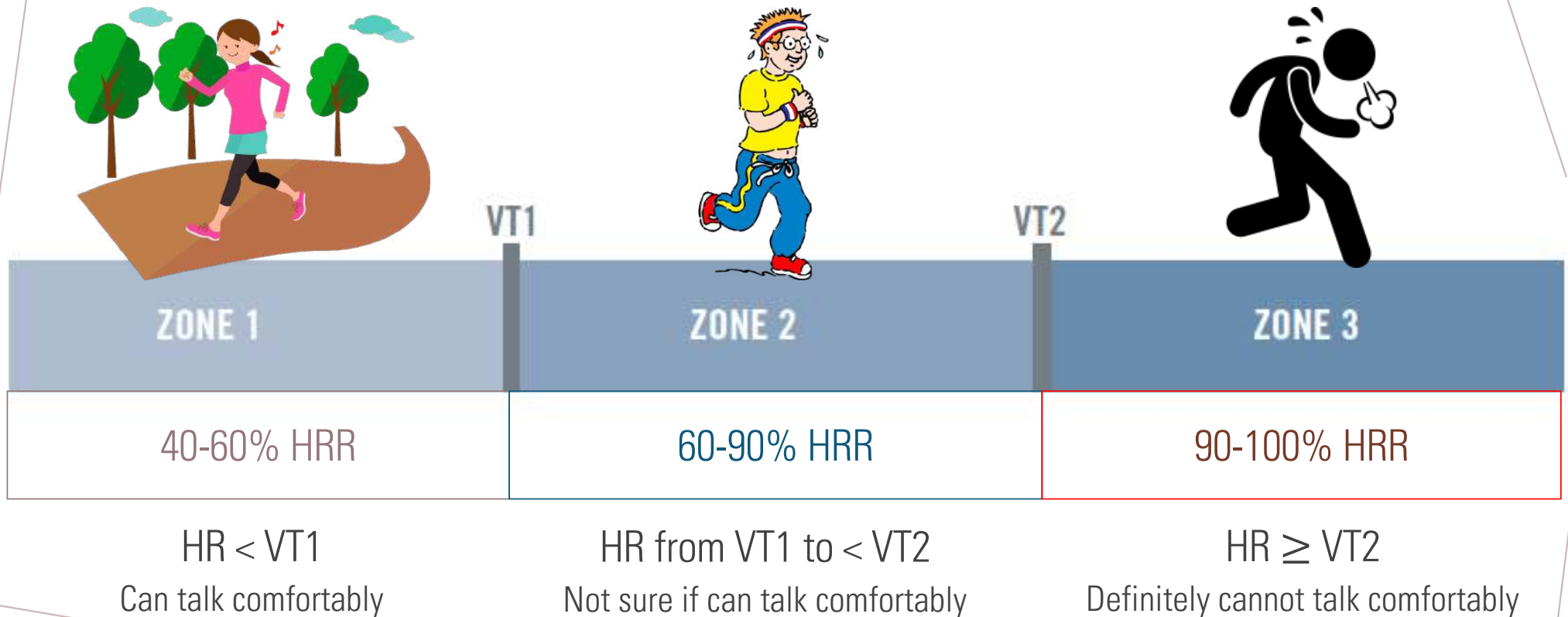
Source from: R. L. Hughson, *Appl. Physiol. Nutr. Metab.*, vol. 34, no. 5, pp. 840–850, Oct. 2009, doi: [10.1139/H09-088](https://doi.org/10.1139/H09-088).

INTENSITY - CALORIC EXPENDITURE

- When the human body burns fuel, O_2 is consumed, which yields calories to perform work.
- The number of calories produced per liter of O_2 consumed varies according to the fuel:
 - 4.69 kcal/L of O_2 for fats
 - 5.05 kcal/L of O_2 for glucose
 - A value of 5 kcal per liter of O_2 is sufficiently accurate



INTENSITY - VT1 AND VT2



Refer to Table 8-11: Three-zone Intensity Model Using Various Intensity Markers

THE THREE-ZONE INTENSITY MODEL

Intensity Markers	Zone 1	Zone 2	Zone 3
Intensity	Moderate	Vigorous	Near max/Max
Metabolic Markers VT1 & VT2 (HR@VT1 & VT2)	<VT1	=VT1 to <VT2	≥VT2
RPE (0-10)	3-4	5-6	7-10
RPE (6-20)	12-13	14-17	≥18
%VO ₂ R	40-60%	60-90%	≥90%
%HRR	40-60%	60-90%	≥90%
%MHR	64-76%	77-95%	≥96%
METs	3-6	6-8.7	≥8.8
%VO ₂ max	46-63%	64-90%	≥91%

Refer to Table 8-11: Three-zone Intensity Model Using Various Intensity Markers

TYPE / MODALITY



TYPE - MACHINE



Treadmill



Elliptical



Rowing ergometer



Arm Ergometers



Recumbent Bike



Stepmill

WATER-BASED EXERCISE

- Reduces orthopedic loading due to buoyancy
- Trains different muscle groups than those used during ambulatory activities
- Energy costs:
 - Water walking/jogging – strongly related to water depth; increases with speed
 - Swimming – highly variable; depends on velocity, stroke, skill, and technique



OTHER TRAINING MODES



Circuit



Mind-body exercise

OUTDOOR EXERCISES & GREEN EXERCISES

Green Exercises:

- Exercises performed in natural environments
- Fishing
- Gardening
- Hunting
- Walking to a store
- Golfing
- Tennis
- Group outdoor class
- Skiing

Running
Cycling
Swimming, aqua-workout
Roller skating, skateboarding
Hiking, trekking
Mt./rock climbing
Kayaking, rafting, boating, canoeing
Surfing, wakeboarding, stand-up paddle
boarding
Skiing, tubing, snowshoeing

EXERCISE PROGRESSION

- Making periodic, progressive increases in exercise frequency, duration, or intensity.
- As a general rule, increases in frequency, duration, or intensity should be limited to **10%**, and increases should be made only after the body has adjusted to the new program.

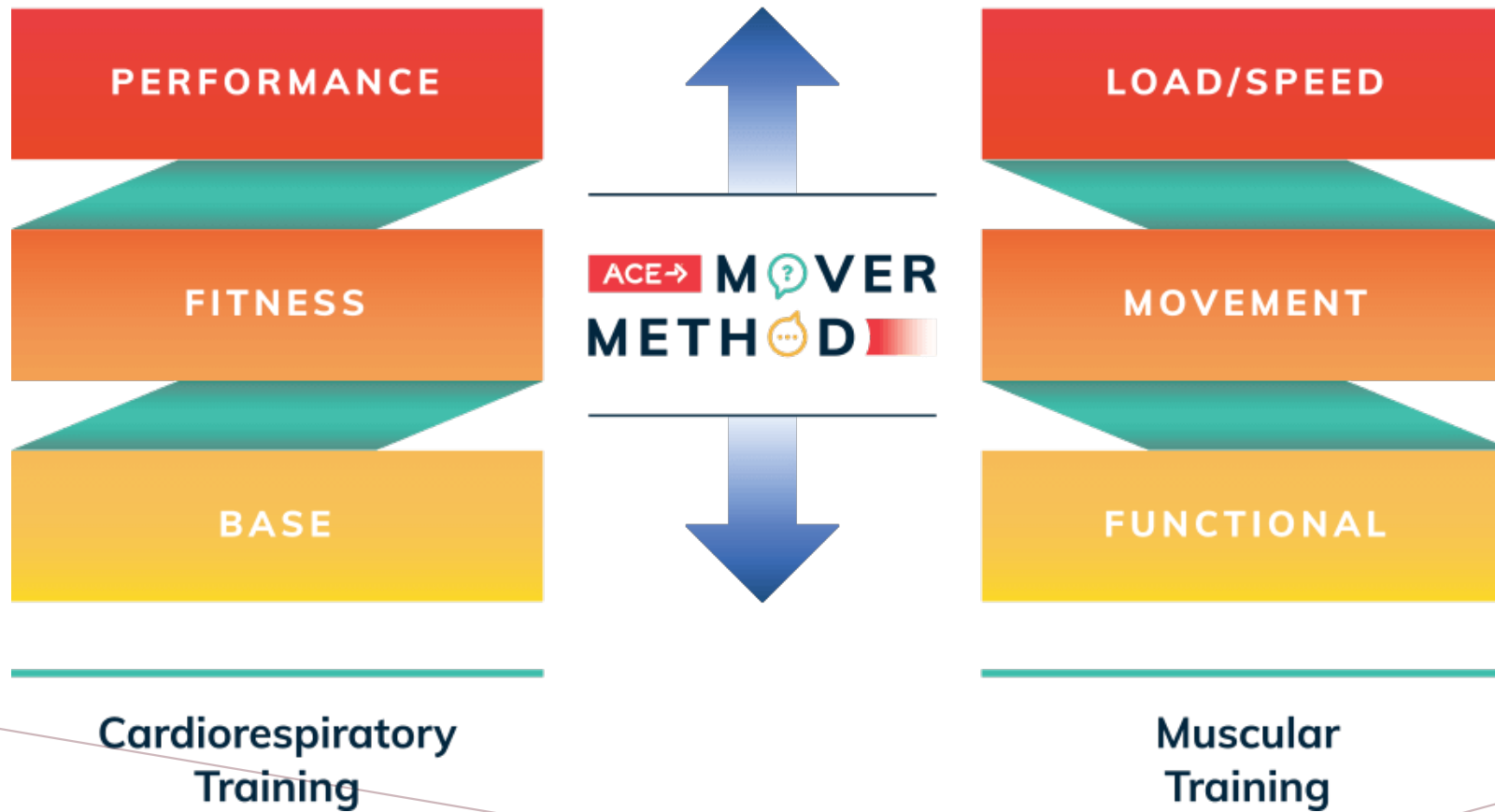


FITT FOR REDUCING SEDENTARY TIME

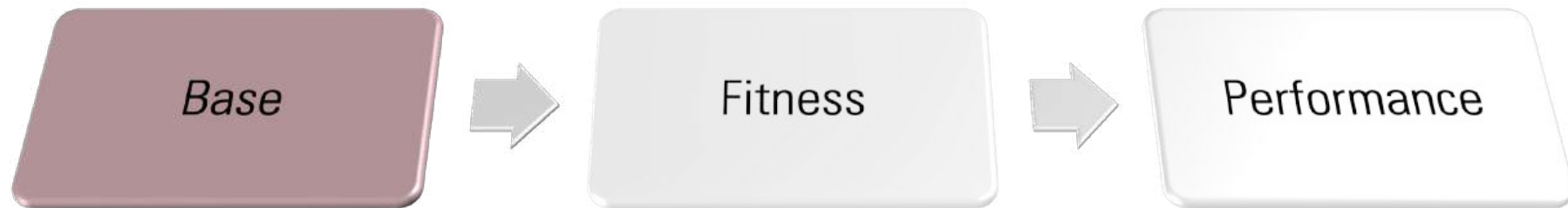
- Frequency
 - Take breaks from sedentary time every 60-120 min
- Intensity
 - Light intensity: <25 to 50% VO_2max , >1.5 to <3.0 METs, or <VT1
- Time
 - Limit overall discretionary sitting time to < 2 h/d
 - Each break lasts 5-10 min
- Type
 - Light activities: e.g. housework, occupational tasks
 - Modify environment to limit sitting, e.g. remove chairs



ACE→ Integrated Fitness Training® Model



ACE IFT MODEL – CARDIORESPIRATORY TRAINING



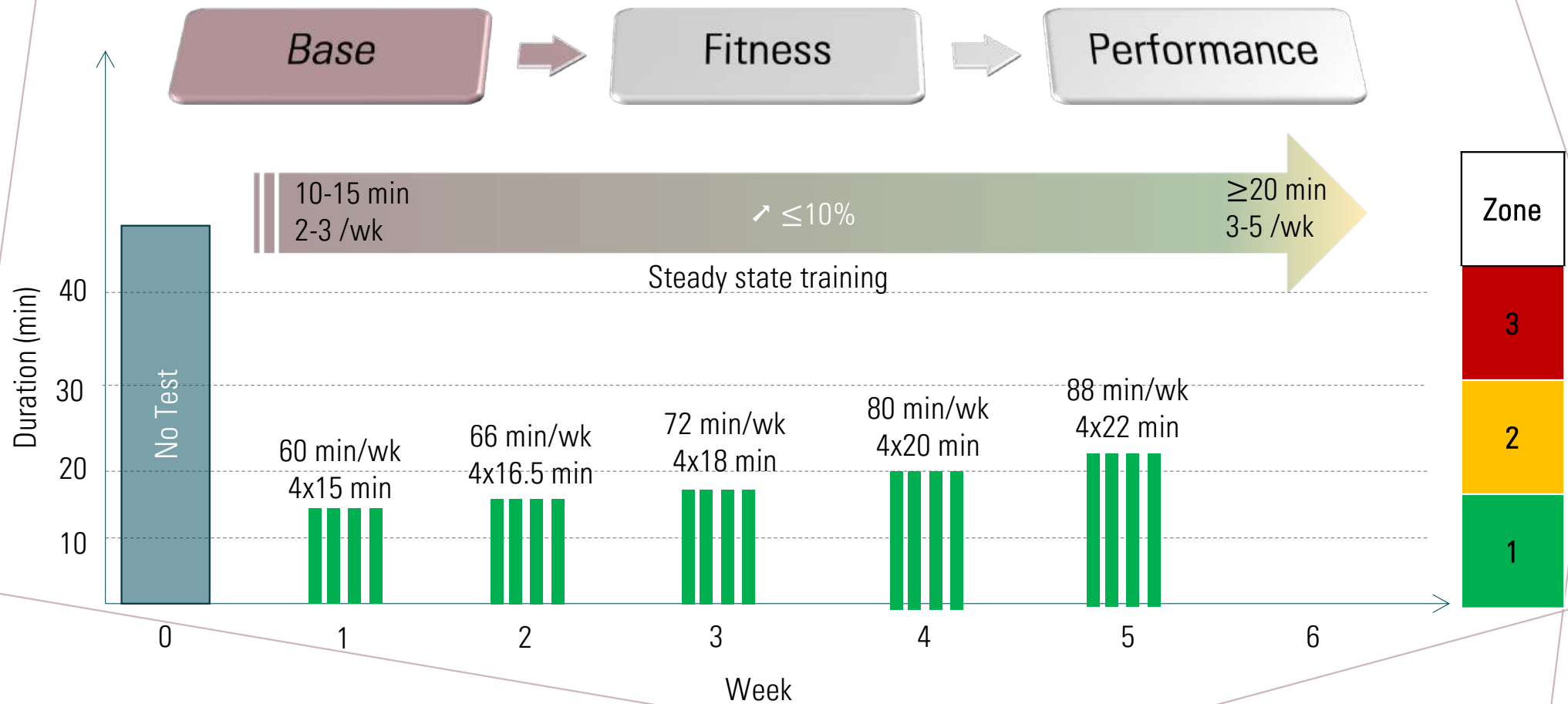
Training Focus:

- Clients: physically inactive individuals
- Goals: 1st, improve health; 2nd, build fitness
- Purpose: start regular CT
- Intensity: low to moderate

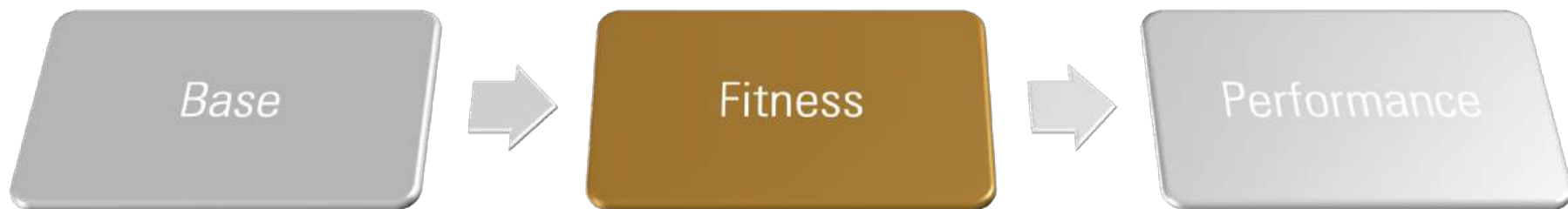
Emphasis

- Positive exercise experience to encourage exercise habit building and increase self-efficacy

ACE IFT MODEL – CARDIORESPIRATORY TRAINING



ACE IFT MODEL – CARDIORESPIRATORY TRAINING



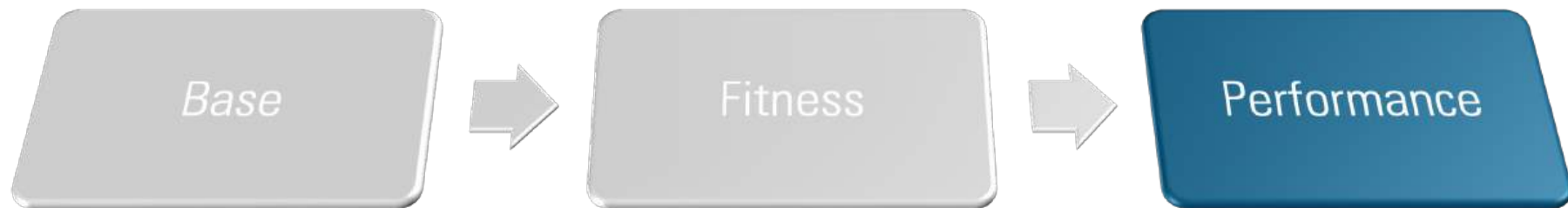
Training Focus:

- Goals:
 - Enhance cardiorespiratory efficiency
 - Improve or maintain fitness and/or health
- Programming: increase the duration, frequency, intensity
- Introduce Zone 2 intervals
 - ↗ ability to utilize fat as fuel just below VT1
 - ↗ CR efficiency at VT1
 - ↗ HR @ VT1
 - Add variety to the program

ACE IFT MODEL – CARDIORESPIRATORY TRAINING



ACE IFT MODEL – CARDIORESPIRATORY TRAINING



Training Focus:

- Clients with endurance-performance objective and/or competitive endurance goals
- Goals:
 - Enhance cardiorespiratory efficiency
 - Build anaerobic capacity
 - Increase anaerobic power (only for fit and competitive clients)
- Programming:
 - Ensure adequate training volume
 - Ensure appropriate training intensity and recovery

ACE IFT MODEL – CARDIORESPIRATORY TRAINING



- Volume
 - **Rule of 3:** Empirical rule from marathon training that weekly training volume should be about 3 times the race duration
- Volume of Intervals
 - Varies according to duration of the event
 - E.g. **1.6 km** event: **(2.5x race distance)** 10x400 m intervals
 - E.g. **10 km** event: **(1x race distance)** 6x1 mile intervals
- Rules are highly empirical
- May fit for serious competitive athletes, but need to scale down for nonathletes

VOLUME MULTIPLIER FOR HARD TRAINING INTERVALS VS EXPECTED RACE DURATION

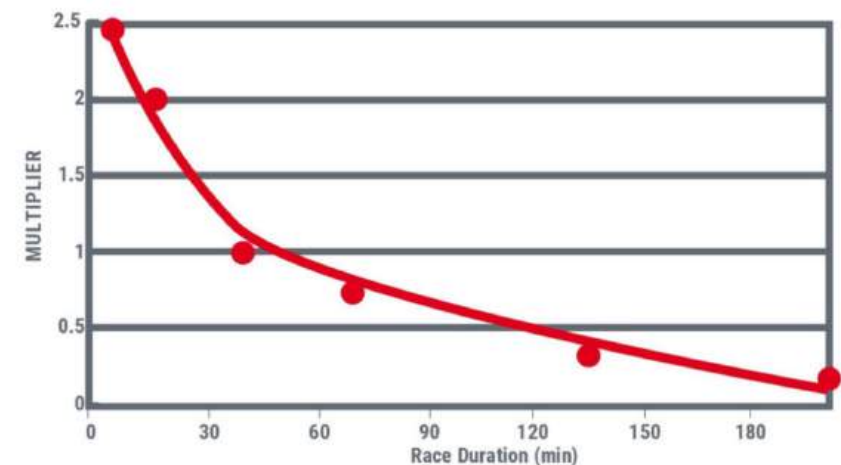
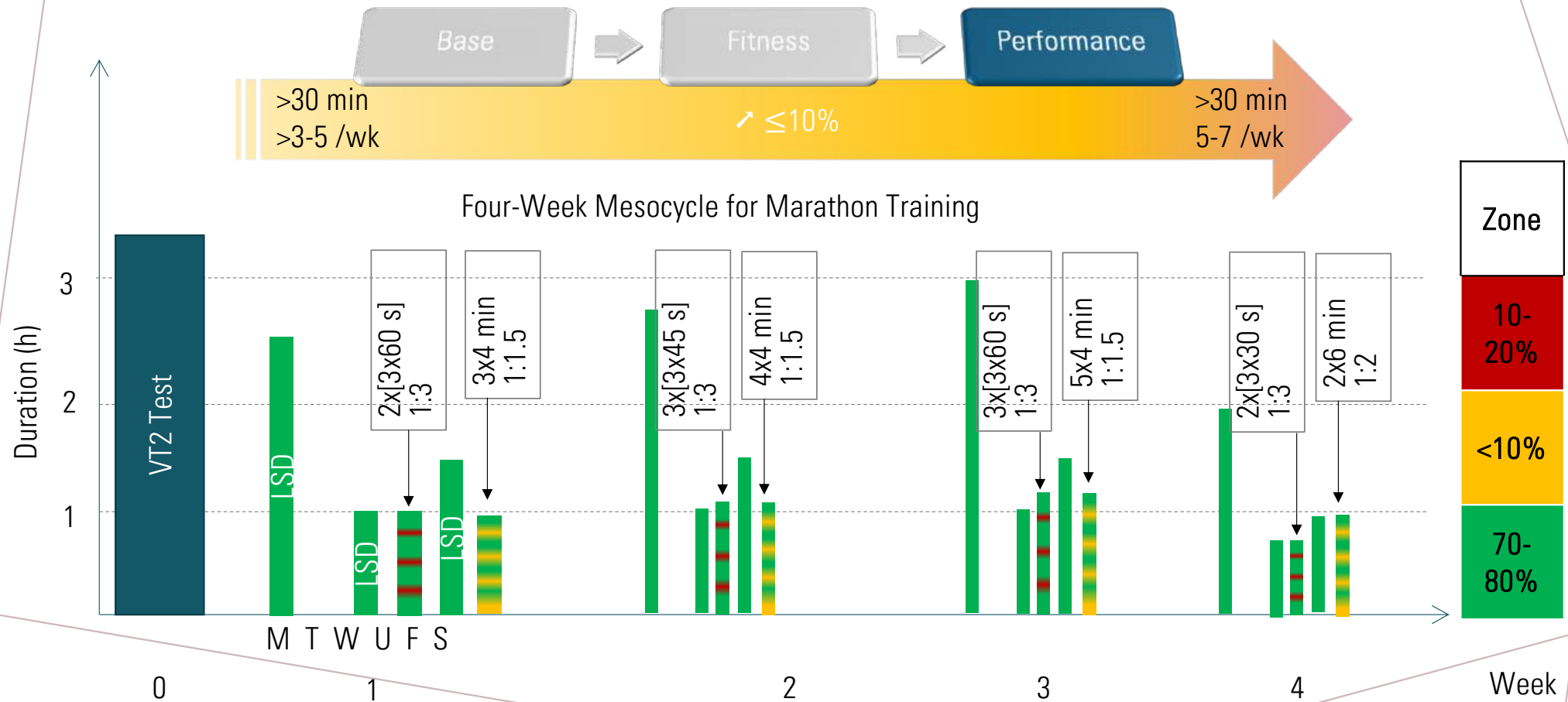


Figure 11-5

ACE IFT MODEL – CARDIORESPIRATORY TRAINING



ANAEROBIC POWER



50s rest
70s, 110% VO ₂ max
50s rest
70s, 110% VO ₂ max
50s rest
70s, 110% VO ₂ max

PREVENT OVERTRAINING

Signs & Symptoms

- A decline in physical performance
- Elevated heart rate and blood lactate levels at a fixed submaximal work rate
- Weight loss
- Sleep disturbance
- Illness
- Irritability, restlessness, excitability, and/or anxiousness
- Loss of motivation and vigor
- Lack of mental concentration
- Loss of joy



ACE-IFT MODEL - CARDIORESPIRATORY TRAINING

CT PHASE	CLIENT EXAMPLES	DURATION	FREQUENCY	TEST
BASE	Previously sedentary individuals	Steady state 5 min → 20 min	2-3/wk → 3-5/wk	None
FITNESS	Regular client to improve/maintain CR fitness; 1-time goal of a 10km competition	>30 min	3-5/wk	HR@VT1
PERFORMAN CE	Qualifying for national even; placing top-5 in a championship	>30 min (activity specific)	5-7/wk	HR@VT2

<
10%

ACE-IFT: CARDIORESPIRATORY TRAINING

CT PHASE	PURPOSE	VOLUME		
		ZONE 1	ZONE 2	ZONE 3
BASE	Improve health and build fitness for previously sedentary individuals	100%		
FITNESS	Improve/maintain CR fitness/health	Varies	Interval @ VT1 ⇒ <VT2	
PERFORMANCE	Enhance CR efficiency; build AN capacity	70-80%	<10%	HIIT 10-20%

<
10%

COMPONENTS OF A CARDIO TRAINING SESSION

CT SESSIONS – STEADY STATE TRAINING

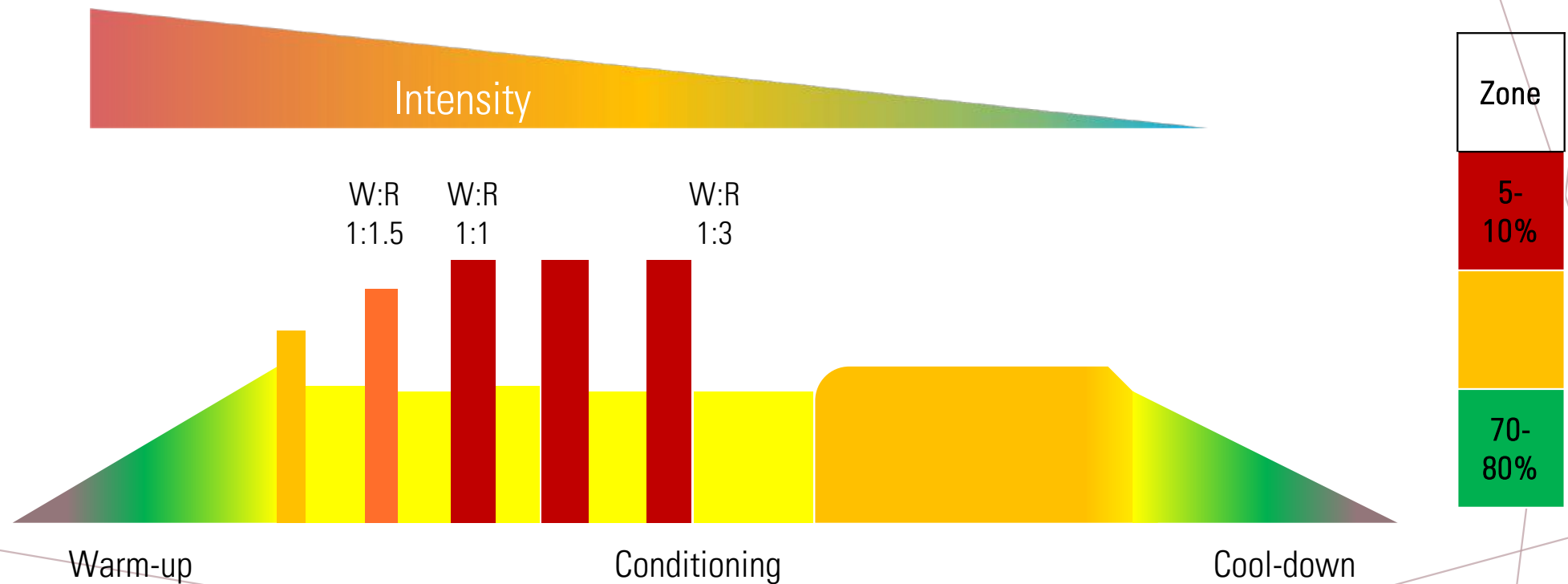


- ↗ Blood flow to target muscles
- ↗ HR to ↘ O₂ debt
- Prepare nervous system
- ↗ Muscle core T to ↗ unloading of O₂ to the muscles



- Prevents blood pooling in extremities
- Remove metabolic wastes
- Stretch to improve flexibility

CT SESSIONS – INTERVAL TRAINING



CARDIORESPIRATORY TRAINING – EXTRA READING MATERIALS

[1] U.S. Dept of Health & Human Services, “Current Physical Activity Guidelines”.
<https://health.gov/our-work/physical-activity/current-guidelines>

[2] K. S. Seiler and G. O. Kjerland, “Quantifying training intensity distribution in elite endurance athletes: is there evidence for an ‘optimal’ distribution?,” *Scand J Med Sci Sports*, vol. 16, no. 1, pp. 49–56, Feb. 2006, doi: [10.1111/j.1600-0838.2004.00418.x](https://doi.org/10.1111/j.1600-0838.2004.00418.x).

EVIDENCE-BASED PRACTICE



EVIDENCE-BASED PRACTICE



EVIDENCE-BASED PRACTICE



Search for bias

Ask yourself, "Does the claimant have anything to gain by promoting this point of view or product?"



Read the research

If research is cited, review the full research paper to get an idea of sample size, study design, conclusions, study limitations, and validity and reliability. This information will shed light on potential logical fallacies. If no research is cited, the claim is likely not an example of evidence-based science.



Evaluate for truth

Signs that something is not true include overstating or overgeneralizing research results, misinterpreting correlations, reliance on anecdotal experience, and selective reporting.



Look for consensus

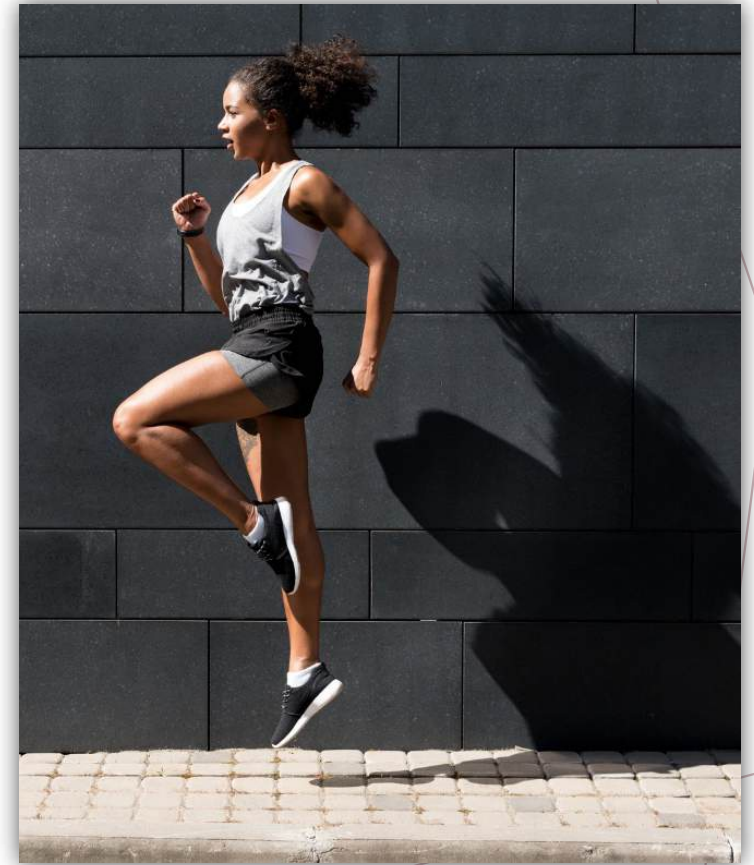
For topics with a public health impact, respected organizations (e.g., the American Heart Association, Centers for Disease Control and Prevention, and World Health Organization) often offer a consensus opinion. These are widely available at no cost.

APPLY EVIDENCE-BASED PRACTICE

Read the statement regarding REHIT on Pg473

Summary:

- HIIT/SIT were not actually time-efficient as famed, with most protocols requiring a time commitment of ~120 min/week (Vollard & Metcalfe, 2017)
- HIIT/SIT have the potential of inducing negative perception
- ACE-sponsored research reports that cardiorespiratory REHIT is an effective, time-efficient, and safe strategy for achieving meaningful health and fitness benefits



APPLY EVIDENCE-BASED PRACTICE - REHIT

1. Check out the referenced journal article
2. Check out the credential of this journal
3. Search for bias
4. Evaluate for truth. Read the article critically. Is the finding significant? Is there over-generalization? Is the sample size sufficient?
 - What is the null hypothesis? Is it meaningful?
5. Looking back at the original statement, what do you think now? How can the research be better conducted?
6. Look for consensus
 - What is the relevant consensus that is widely available at no cost?

